

Alt(ernate) CHI: Using Alternate History Artifacts in Research

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Abstract

The CHI community continually looks to new horizons as we strive to innovate, and central amongst our goals should be a socio-technical future that is demonstrably better than the past. Accordingly, whilst we may look to the past to reflect on our own trajectories as a field, there remains unfulfilled potential in using the lens of history to its fullest capacity. We consider there are opportunities to facilitate research through design not only using our own histories but histories which run counter to our reality. In this paper, we call for the use of alternate histories as provocative design resources in HCI and CSCW research with participants, showing how we leveraged alternate history to explore contemporary matters around public service media and presenting guidance for other researchers to use this approach.

CCS Concepts

• **Human-centered computing** → Human computer interaction (HCI); HCI design and evaluation methods; User studies; Human computer interaction (HCI); HCI theory, concepts and models.

Keywords

Alternate History, Design Fiction, Provocative Prototyping, Counterfactual Artifacts

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1 Introduction

The HCI and CSCW communities continually looks to new horizons as we strive to innovate, and central amongst our goals should be a socio-technical future that is demonstrably better than the past. Accordingly, whilst we may look to the past to reflect on our own trajectories as a field [42], there remains unfulfilled potential in using the lens of history to its fullest capacity [4, 18, 54]. However, some researchers have used the past in innovative ways, leveraging history as a powerful resource, either using it as inspiration for future designs [4, 5, 18, 54, 55, 63] or opening the past as a malleable design space [26, 56]. Many researchers recognize the role that historical perspectives can have in our work.

The past is not only a plentiful source of inspiration for those willing to use it but a record that can be manipulated by those with the power to do so for their own gain. The phrase “those who cannot remember the past are condemned to repeat it” [52], has become a rallying call to remember the mistakes of the past, but which pasts are we remembering and what learning opportunities are we missing? While there is a multiplicity of interpretations of history, we are limited to interpretations of what could have conceivably led to our current world. In this paper, we consider there are opportunities to facilitate research through design not only using our own histories but histories which run counter to our reality. We call for the use of alternate histories as provocative design resources in HCI and CSCW research with participants, showing how we leveraged alternate history to explore contemporary matters around public service media.

Alternate history is known by many names, including counterfactual history, allohistory, parahistory, uchronia and alternate

world history. Outside of HCI scholarship and design practice, there is a long tradition of alternate history media. Many of these works can be found in entertainment, with examples like the novel *The Man in the High Castle* [16], which like many alternate histories that followed, explored an alternative outcome of the Second World War. Significant events such as wars may provide points of departure from our reality, but much like future-oriented fiction, there are a multitude of variations incorporating changes on scales from world-altering to highly personal. Speculation over alternate history has also become the focus of several online groups, such as the subreddit *r/AlternateHistory* [65] and alternate history wikis (e.g. [66, 67]). These communities feature discussions and compilations of counterfactual narratives. Conversations include members debating the impacts of changes, the feasibility of different alternate history narratives, and opinions on historical fiction.

In this research we benefited from a distinctive community of our own: our research team includes an HCI researcher with knowledge and experience of an alternative version of public service media (PSM). He has experienced much of this first-hand, as a resident of an alternate timeline.

When the second author and colleagues established communication with the rest of our team around six months prior to the writing of this paper (by means of a ‘multiversal communication device’ created by researchers at University College Newcastle) we discovered a similarity between the research topics of the first and second authors and decided a collaboration would be beneficial. After some infrastructure setup at Newcastle University, communication continued using the platform “BBC Social” - a popular social media service in the corresponding universe - and provided an opportunity for the first author to explore the other universe. Following much deliberation, we have agreed to call any universe we publish work in “Universe α ” and the corresponding alternative timeline “Universe β ”. While this may not be ideal, we have made this decision for the sake of brevity given there is no standardized naming convention. When we seek to publish a companion paper to this in Universe β , we will reverse the naming convention so it is named Universe α as it will be the publishing universe.

Although it is not yet common knowledge that other timelines exist, we are trusting the HCI and CSCW communities with this information early as we believe it presents excellent research opportunities and we hope this will be just one of many research projects leveraging what would have once been dismissed as fictional counterfactual histories [2, 15, 21, 26, 39].

2 Related Work

2.1 Design Fiction

Design fiction is a design and creation practice drawing on fictional and speculative narratives or settings. It has been used to explore a broad range of futures, including speculation over performing arts [24], voice assistants [50], privacy [59], and the environment [49, 61]. As well as these portrayals of singular futures, *Contravisions* [48] shows how multiple alternate outcomes to technology interventions can be explored.

In another example of a forward-looking design fiction, Sturdee et al. [57] demonstrate how design fiction can take explicit inspiration from existing works of fiction, with their use of the cyberpunk

world of *Bladerunner* [64]. Science fiction has been somewhat explored in HCI [19, 32, 62], although Jordan and Silva [34] contended in 2021 that science fiction is an untapped opportunity. The online communities that form around works of fiction like these have also been explored for sourcing feedback over speculative works and the ideas that researchers can embed within them [13].

Design fictions have been integrated not only into research as in the previous examples but in papers themselves, such as Mitchell et al. [40] incorporating existing designs examples into an overarching narrative structure. Through fictional abstracts, researchers have also looked not sideways into other timelines but forward to potential futures for the CHI community in work by Baumer et al. [3], to consider the cultural, political and social changes that could influence our work. Kirman et al. [36] took the approach to the further extent of creating a fictional conference through which authors could submit imaginary papers. *A Mulching Proposal* by Keyes et al. [35] and *CHI and the Future Robot Enslavement of Humankind* by Kirman et al. [37] illustrate how satirical future-oriented design fictions can provide a humorous reflection on our own ethical frameworks and research directions (as well as a reminder of the worrying rumors we hear about universes λ and μ). Dragicevic et al. [20] gave us hope that other researchers had also collaborated with other realities to run an HCI experiment in parallel universes, but this too proved to be a fictional setup, albeit with the experiment simulated in our universe due to their difficulty communicating across the multiverse.

In contrast to relatively realistic science-backed approaches taken in some projects, researchers such as Encinas and Blythe [25] have demonstrated how a more magical approach can be used, although they dispute the ‘fictional’ aspect of this design. The level of realism designers embed within design fictions has been interrogated by Coulton et al. [12], noting the potential pitfalls of an audience mistaking fiction for reality. Some forms of design fiction such as steampunk, as investigated by Tanenbaum et al. [58], are explicitly built around what may be unrealistic design choices. A lack of realism may also be an asset. In *Making by Making Strange* [4], the utility of unusual designs that shun what we might consider realistic are shown to have the potential to defamiliarize everyday technologies.

2.2 Provocative Prototyping for Research

Provocative prototyping, also known as provotyping [41], has developed as a way to use prototypes to elicit responses in an audience, such as participants in a workshop setting. Boer and Donovan in *Provotypes for Participatory Innovation* [6] provided a set of characteristics of provocative prototypes and guidelines for using them. Vines in *Playing with provocations* [60] claims that provocations can “promote critical engagement and reactions from participants, including around subject matters that are hard to articulate or envision”, as well as around taboo topics. As Bruun et al. [8] demonstrate, provotypes can be effective tools for challenging norms and uncovering social tensions.

Provocative prototypes may provide unique functionality but can also involve the removal or reduction of functionality, as demonstrated by Pierce and Paulos’ *Counterfunctional Things* [45]. These objects and their variations [46] present limitations in the face of

the ever-expanding limits of digital technologies, such as low resolution displays and inaccessible cameras. As such, they embody some of the limitations that were present in past technologies to provoke reflection and explore the effects of technological absence. Of related note is Elsdén et al.'s *On Speculative Enactments* [23], which purposively explores the use of design provocations in contexts holding real social consequences, to help further ground the interpretation of provocations.

2.3 Counterfactual History

Outside the immediate sphere of HCI literature, Dunne and Raby's *Speculative Everything* [21] promotes alternate history as a tool for speculation and understanding the importance of historical events. In contrast, in *Why Do We Ask "What If?"* [51], Rosenfeld argues we use alternate history narratives instead primarily as a reflection of our views on the present. We see this in design works that have used alternate history such as *A New Scottish Enlightenment* [1] and *The Golden Institute* [47] that both explore an issue that remains contemporary - sustainable and equitable energy generation - among commentaries on other related topics such as the political backdrops to these developments.

Within HCI and CSCW, Soro et al.'s *Designing the Past* [56] highlights the multiplicity of co-existing narratives about the past and a lack of a single shared history, but instead a malleable design space that is constantly reinterpreted and rewritten, including by our own biases. This is echoed by Chowdhury et al. in their research into memory in migrant families that highlights both what can be missed by mainstream sources and the subjective nature of memories [10].

There have been multiple calls to include a more historical perspective in HCI. DiSalvo [18] spoke of a need for more design history in HCI to help inform the designs of the future by learning from what has come before, which was complemented by Soden et al.'s goal to build a website for related resources in HCI and CSCW [55]. *Time for Historicism in CSCW: An Invitation* [54] went further in providing an overview of CSCW strands and how a historical sensibility could benefit CSCW research.

Beyond this conceptual work, there have been some applications of historical perspectives. In *Historically Informed HCI* [39], Mensonge shows how deliberate use of anachronisms (objects or events shown at times they could not have existed) can highlight parallels between the past and present. Howell et al. [31] demonstrates how the parallel presents that stem from alternate histories can also provide inspiration for speculating over preferred versions of the present day. Using generative AI, Dunnell et al. [22] subsequently enabled people to produce videos of preferred alternate realities centering around Open AI and asked participants how this made them question history and society.

A more explicit example of alternate histories being used in a research project is seen in Forlano and Halpern's *Speculative Histories, Just Futures* [26] through one of their counterfactual actions workshops. The *Reimagining Work* workshop described in the paper used alternate history to explore the history of work and alternative possible futures. Participants moved through historical periods as a game, starting in ancient Greece and Egypt and looking ahead to produce their own counterfactual events. This contrasts with

our proposal of engaging participants with provocations set in an alternate history and demonstrates an alternative way of using alternate history in a research project.

3 Alternate History in Practice

To implement alternate history for research, we designed and implemented two workshops lasting two hours each using three artifacts from the other timeline (Universe β) as provocations. In total, we engaged 17 participants in the workshops. The purpose of these workshops was to understand the types of relationships that participants would like to see between their communities and the public service media organizations (those with a commitment to serve the public interest). The context of public service media in the UK is one in which multiple organizations serve the whole country (such as Channel 4 and the BBC) alongside others that serve individual nations (like S4C for Wales). As an organization with a long history and that is simultaneously international but with strong associations with the UK as a nation, the BBC in particular means a lot of different things to people across Universe α , both positive and negative. In the following subsections we describe how we have applied alternate history to the research project, including our rationale behind why we chose the provocations we sourced from Universe β .

3.1 Alternate Reality Workshops

Given the origin of the provocations we used in the workshops and a lack of public knowledge of other universes beyond our own, we were faced with a difficult decision about how to present the artifacts. Since the workshop participants would likely not believe where we had found the designs, we decided the best course of action was to indicate we had invented them as part of a fictional counterfactual history. As such, we were unfortunately unable to have participants be 'in' on the activity as researchers such as Elsdén et al. [23] advocated for in their own work with diegetic prototypes¹.

To build a set of activities around our three provocations, we used the Civic Probes method [33] as it allowed us to forefront the artifacts as probes alongside activities that leverage the qualities of the artifacts. While the first implementation of Civic Probes by Johnson and Vlachokyriakos sought to use the designers' knowledge and expertise, we were able to position the artifacts as sources of knowledge. We were therefore probing our reality using the artifacts of another reality. This provided learning opportunities and stimuli for discussions that "introduce critical questions and skepticism early" [33] in our overarching research project into public service media. As this form of media has existed for many decades and can be present throughout a person's life starting from a young age, we hoped it would provide many interesting opportunities when using alternate history.

We first carried out a pilot of the workshop using members of our research group in Universe α . Following this, we then ran the workshop with ten members of the public aged 20 to 32 from the North East region of England to investigate their opinions and preferences

¹We consider that others in our search of existing literature may have extended this level of secrecy to their publications out of concern their work might not be published if they were not believed. This could mean there are potentially other inter-universe collaborations that have gone unreported.



Figure 1: Stills from an instructional video on the features of Social on Ceefax from Universe β .

about relationships with public service media. We split the participants into three groups, running concurrently, with one group for each provocation and set of activities. After they had engaged with the activities on their tables, we moved the participants groups between the tables so by the end of the workshop, all participants had engaged with each of the alternate history provotypes.

3.2 Alternate Reality Provocations

3.2.1 Social on Ceefax. In both histories, the BBC (British Broadcasting Corporation) launched a service using Teletext technology in 1974 called Ceefax. This service allowed audiences in the UK to view text-based pages with limited graphics on Teletext-capable TVs using a terrestrial signal. Both universes may have named the service in reference to letting users “see facts” [44], but the development team in Universe β took it in a different direction after launch, with more of an emphasis on community events and opinions than their counterparts in Universe α , who used it for purposes including news, sports results, weather forecasts, TV listings and games. Both services were eventually shut down in 2012, having been replaced by newer technologies [44].

The second author (Universe β) retrieved an instructional video about the provision provided by the BBC Social division on the service (Figure 1). From our research, we have found no evidence of the ‘Social’ division existing in Universe α . As we only had reference images for the service, we decided to recreate the pictures on the screens and display them on a similar television in the workshop using a Raspberry Pi² plugged into a HDMI converter.

We selected Ceefax to be part of the workshop in part for its limitations. The restricted number of lines per page and number of pages contrast with the seemingly limitless capacity of modern social media platforms for new content. This provided an opportunity to use these limitations to reflect on the nature of current media technology. Since the restrictions on the amount of content meant a selection process would have to be used to decide what made it onto the platform, this also gave a chance to discuss who

should be making such decisions in relationships with the public and how they should be making them.

3.2.2 Mixtaper. This artifact notably does not exist in any form in Universe α but shares some similarities to the visual language used by the BBC’s streaming service in both timelines, iPlayer, from when Mixtaper was first released in 2008. Mixtaper was created by Digital UK, a collaboration between the UK’s public service media organizations that also existed in Universe α and helped promote the switch from analog to digital television [17]. Mixtaper allowed users to re-edit segments of shows to produce their own “Mixtapes” of the media they thought should go together, and then share a link to this remix with other people.

The image in the left of Figure 2 was found online in Universe β shows Mixtaper in use just over 3 months after its August 2008 launch. As the image on the screen is somewhat difficult to see, we recreated the interface to the best of our ability (Figure 2) using archival material as a reference. If we had not sourced the artifact from an alternate reality, we could have perhaps created a prototype using an interface design application and then displayed it on a computer monitor suitable for the era to simulate the appearance of the time. Fortunately, that was not necessary.

We selected Mixtaper to use as a provocation in our research because its position as a participatory media experience that was not entirely user-generated but instead used media owned by the public service media organizations created tensions we could explore with participants. Due to the features around sharing creations between users, this also facilitated conversations about the types of connections people could make or maintain using participatory media.

3.2.3 Community Programme Unit. The third artifact is a video featuring a compilation of footage relating to the Community Programme Unit, or CPU, which was a division of the BBC in both timelines and remains one in Universe β , while its last production aired in 2002 in Universe α [30]. The CPU focused on empowering people to make their own media, beginning with television shows

²For Universe β , see Cherry Pi.

³Sanger Shepherd ceased operations in Universe α in 1927, indicating the point our histories diverged predates this.

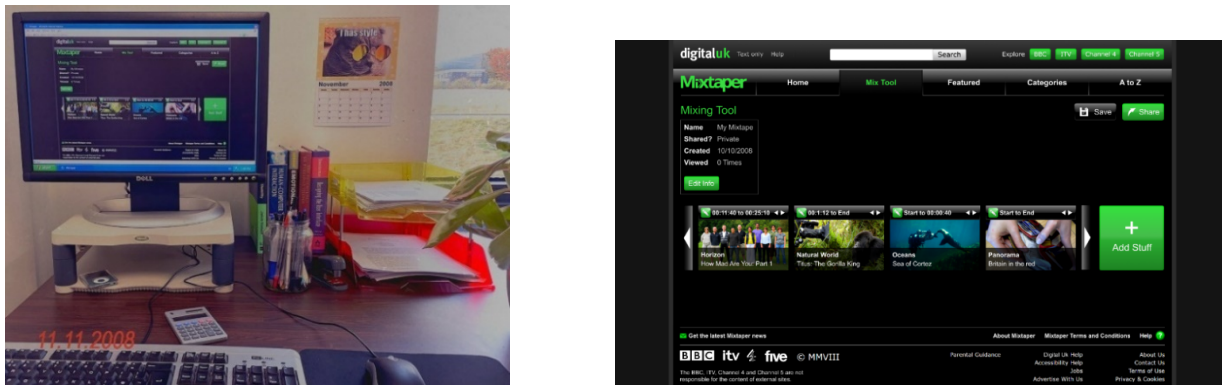


Figure 2: Left: Scan of a photo from Universe β showing the mixtape editing page of Mixtaper in use, taken on a Sanger Shepherd T760 camera³. Right: Our recreation of the 2008 Mixtaper UI from Universe β showing the “Mixing Tool” page in which users could edit media together from the major UK public service media organizations.



Figure 3: Images from the compilation of footage created by the Community Programme Unit, compiled by the BBC Social division of Universe β .

though a production called Open Door. This format allowed people to produce their own short films about issues they cared about using the corporation’s equipment. Concepts were first vetted by the CPU before being put into production, with examples including shows from campaign groups on issues like racism in media and nuclear disarmament, as well as community groups like the Women’s Institute [43]. The format was followed up by other similar initiatives, but the CPU eventually moved onto a video diary model by 1990 with the advent of higher quality home video equipment. In Universe β , the CPU has continued into the present day under the guidance of the same ‘Social’ division, that appears to have created Social on Ceefax and operates a social media platform of the same name. As the division seems to exist exclusively in Universe β , it may have been a catalyst for many of the changes we have found.

Thanks to our connection with Universe β , we did not have to edit an already published compilation down to a suitable length for the workshop or apply the branding of an apparently fictional ‘Social’ division. We instead downloaded the video (Figure 3) directly from Universe β (due to low inter-universal bandwidth making streaming unreliable) and played it to our participants before they engaged with the activities associated with it.

We chose this artifact as a provocation because it allowed us to put participants in the position of making the kinds of decisions the CPU’s organizers made in selecting who was given a show, exploring their beliefs about who could be supported by PSMs. By withholding what the CPU is currently doing in its continued existence in Universe β , we could also use this ambiguity to encourage participants to consider what a group that had once focused so heavily on broadcast and online video media could still be doing to support people in a time when the online streaming of media created by the public had proliferated. We encourage other researchers to also consider how selectively withholding certain information about their alternate histories could lead to intrigue and speculation.

4 Applying Alternate History

The authors were fortunate to encounter an alternate reality with differences from our own that aligned with our research interests. However, we appreciate that, sadly, many researchers will not have the same opportunity and instead be restricted to the reality of their own universe. In the following section we apply what we have learned from this experience in dialogue with existing related work

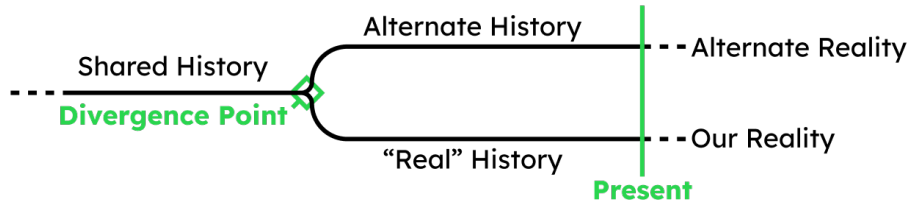


Figure 4: Diagram of an alternate reality branching from our reality in a single divergence point.

to inform others in producing their own fictional alternate histories in the absence of a pre-existing alternate reality they can use.

Firstly, researchers should consider whether using an alternate history approach to provotyping is the best approach, given the specific characteristics of their research objectives. It is understood that taking a historical perspective can allow freedom from established norms and explore apocryphal or unrealized possibilities as a reflection on our own reality [56] but we have found alternate history, by its nature, means considering the impossible (at least for our reality). Alternate histories are distinct from so-called “secret histories” [15], which could have conceivably happened in a broadly accepted version of our history, even if we don’t have the facts to confirm or deny if they did. For example, a story set in the past with characters that do not influence the wider historical context or an artifact that could have existed without being documented until now could be part of a secret history. Meanwhile, events that conflict with what the observer considers to be our actual history would be considered as part of an alternate history as they are “counterfactual” [2, 21].

Researchers may wish to first design the alternate history in which their artifact is situated or begin with the artifact and then build the alternate history around the artifacts. Such an artifact-first approach could allow for initially disconnected provocations to be connected under and overarching theme and narrative, as has been demonstrated by future-oriented design fiction [40]. Alternatively, the artifacts and history could shape each other as the researchers design what best suits their research goals and the audience for their provocations.

4.1 Points of Divergence

Many alternate histories are framed as the result of a single divergence point, in which something changed from our reality, with everything that happens beyond that being a result of this change [15], leading to an alternate history (Figure 4). Shippey [53] argues a point of divergence should aim to fit criteria to be effective, namely that it should be “(1) plausible, (2) definite, (3) small in itself and (4) massive in consequence”, but not all works follow this structure and some forego a clear divergence point altogether or have multiple divergences that combine to create the alternate history [15]. For instance, we determined that putting time into discovering the divergence point between the realities of the co-authors was superfluous as it was outside of the scope of our research. Rather than distract the participants in our research with an (alternate) history lesson, we instead focused on the artifacts as they related to the purpose of the study.

Following the divergence, the resulting timeline can lead to utopias, dystopias, and anything between. The choice of how utopian or dystopian the designer wishes to be with their speculation depends on the purpose of the provocations. For example, the future-oriented project *Blackbox* [59] uses a dystopian approach to explore an “undesirable and frightening” possibility for privacy. Artifacts that present a more positive course of past events will reflect badly upon our reality by comparison, whereas more negative alternate worlds show that it could have been worse for us. However, such ideas are very subjective and not all interpreters of alternate histories will view them in the same light as their creators, given there are already so many understandings of our own reality and history [11, 56]. While the second author has complaints about his universe, we don’t believe that Universe β should be considered utopian or dystopian relative to Universe α , merely different from it. This was useful for our research project because it allowed us to engage participants with other possibilities and discover their perspectives in a nuanced process, using ambiguity as a resource [27], rather than presenting something that was clearly meant to have one interpretation. As Gaver et al. [27] argue, ambiguity can be powerful for raising topics without imposing answers.

4.2 Implications of Aesthetics and Nostalgia

When designing alternate history artifacts, the setting can provide more than a social and technical backdrop against which to frame the artifacts. Trends like cottagecore [7] and vaporwave [29] that are rooted in the visual and musical associations of eras and ways of life demonstrate the power of aesthetics in eliciting engagement through the large communities that developed around them. When situating an artifact in the past, researchers should consider the aesthetic elements that could enhance it. Aesthetics can produce a more engaging experience and convey meaning when used in future-oriented design fiction [9]. A historical sensibility in design allows researchers to additionally use aesthetics to elicit associations with the past, such as invoking nostalgia for a time period. The use of nostalgia as a heavy-handed tool for attention has rightfully been questioned and parodied [38]. However, we see the potential for researchers to leverage nostalgic elements in designs to put into focus those aspects of how we engage with artifacts to open them up for questioning rather than shunning them. In our research, we used provocations that had aesthetic similarities to artifacts in the universe in which the participants originated, aware this could make provocations more engaging for those with connected lived experiences. By making activities that specifically involved what could be seen as flaws and limitations of those artifacts, we were able to probe what made them appealing as a learning opportunity.

4.3 Realism and Immersion

Incorporating aesthetic qualities associated with the time in which artifacts are set could also make them more believable, contributing to a feeling of immersion for those who engage with them. In our research we know our artifacts are possible since they came into being in their own universe. Researchers without this reassurance who need to create their own artifacts may wish to investigate so-called ‘real-world’ artifacts for inspiration and familiarize themselves with details that could anchor their own artifacts. For example, if an alternate history involved no significant differences in technological capability from their history, researchers could design artifacts that incorporate the limitations of the time.

Contrary to this realism, researchers may instead wish to break away from such limitations or use particularly stylized designs even if it comes at the expense of believability. The proliferation of the steampunk genre of fiction and its related design sensibilities have demonstrated how counterfactual objects can be detached from a time period and still provide an engaging experience. While steampunk often takes inspiration from Victorian aesthetics and can be set in the period, it combines these with technologies that appear beyond the capabilities of the time [14, 58]. As Tanenbaum et al. [58] indicate, the conflicting worlds of steampunk featuring advanced technology and the oppressive societal norms associated with its historical aesthetic elements can be used to explore social issues like our own norms and values. Researchers may similarly wish to incorporate clashing features into their artifacts to this end.

Future-oriented speculation, argue Dourish and Bell, “is inherently grounded in assumptions about the problems and opportunities of the time at which it is written” [19]. While this means such fiction is likely not an accurate portrayal of the future, for researchers it provides an excellent opportunity to embed current anxieties and topical matters into design fictions. When developing an alternate history, researchers can pull from both the issues of their time and those from our own historical records that the alternate history intersects with. We see this with the Golden Institute [47] in which the overarching theme resonates with contemporary issues around renewable energy while the content of the work references the tensions the characters within it would have felt between the US and the Soviet Union. We encourage designers to consider the purpose of incorporating historical anxieties into the work. The provocations and activities we used in our research focused more on the relationships between media organizations and the public and limitations of the technologies we found in the alternate history reflect on those in the participants’ reality. We did not incorporate the international social or political landscapes of the time explicitly in the parts of the alternate history we showed since this was out of the scope of the discussions we were interested in for our research.

4.4 Biases and Anachronisms

Designers of alternate histories should be aware of the dangers of embedding their own biases within their designed worlds. If the authors of this publication had needed to create the artifacts in this paper, they would have had to reflect on their own positions not only as academics from western backgrounds with their own lived experiences, privileges, and biases, but as residents of a certain

time. As the co-author of the alternate history story *The Difference Engine*, William Gibson, said “We don’t see the Victorians as they saw themselves, at all, nor as the inhabitants of our own future will see them” [15]. Our lack of lived experience of events and people before our time may bias our views and make it far too easy to create a portrayal of people in the past that implies they were, among other things, as simplistic and ignorant as the portrayal itself.

While it is entirely possible to design an alternate history that diverges so far from our own that the people within them are so far removed in their culture that there is little of our world to misrepresent, much alternate history takes place in timelines relatively close to our reality. In such cases, to counter the possibility of disrespectful misrepresentation, we urge designers to research any time periods, people, and individuals that their work intersects with and empirically ground their representations. Robust research may also help lend believability to artifacts by helping ensure there are no immersion-breaking anachronisms if that is the researcher’s goal. The question of maintaining a sense of realism within the artifacts and their surrounding alternate history may seem unusual given we are discussing things that by their nature are counterfactual. As Dunne and Raby argue in *Speculative Everything* [21], you can ask an audience to believe something inside an imagined world that might be impossible, but anything improbable could cause the concept to fall apart, like for example, if people on their way to Mars in a story didn’t react the way we would expect to an alien ship appearing.

The audience for the artifacts may also influence the degree to which any historical anachronisms affect their engagement with them, especially if they have lived experience of times the provocations are set in or related real-life artifacts. While some participants may not become distracted by aspects of the artifacts or alternate history they find unrealistic, in the words of the second author during our initial correspondence, “it is better to have them focused on what you want to research and not why Winston Churchill is wearing a smartwatch even though your research is about someone else being first to reach the south pole.”

Considering the audience for your provocations will also help gauge how they may react to historical touchpoints. As we previously mentioned, historical artifacts have the potential to elicit feelings of nostalgia that you can either use directly or subvert for questioning. It is important to consider whether the designs of provocations and the period they are set in allow you to use such an approach to resonate with people or whether your choices in designing the artifacts would not elicit nostalgic responses, perhaps because of a lack of lived experiences with the touchpoints or period you are referencing. Researchers may also wish to deliberately reduce the chance of nostalgic engagement if this is outside of the scope of their research and like an anachronism might do, it could distract from the purpose of the provocations.

5 Discussion and Conclusion

In this paper we have presented our proposal for using alternate history as the basis for producing design fiction artifacts to be used as provocations in research settings. We have demonstrated how we implemented artifacts from an alternate history as provocations in



Figure 5: This research is a cross-institutional and cross-universal collaboration between researchers at parallel institutions. Left: The logo of University College Newcastle of Universe β . Right: The logo of its counterpart, Newcastle University of Universe α .

our research into public service media through our cross-universe collaboration (Figure 5).

Our work is an attempt to bridge the gap between historical perspectives in HCI and design fiction. This method is an answer to the invitation for more historicism in CSCW by Soden et al., providing a way in which researchers can engage in design with “resources that can inspire design, support comparative analysis, and develop a deeper understanding of technology development and its social consequences” [54] in HCI and CSCW research through design. Through this we contribute a movement of the scope for provocative prototyping into the past.

This alternate history approach contrasts existing perspectives on provocative design like counterfunctional objects [45, 46]. Whereas counterfunctionals and other objects like the drift table [28] limit the functionality of contemporary objects, alternate history artifacts can embed within themselves inherent technological limitations as a consequence of their position within time. This provides a grounding of the limitations and a surrounding historical context with which to frame the provocations.

Our approach provides a means by which researchers can present counterfactual historical concepts to research participants in a digestible format that means external information can be optional, as concepts can be embedded within artifacts. This relates to a tension we found in Forlano and Halpern’s prior work using alternate history in HCI [26] in which participants imagined their own counterfactual historical events using a game. Some knowledge of the past was required to engage but in their own words, they did not want to spend time “educating” participants, and so found a solution in giving some information and relying on participants’ own knowledge and imaginations. We hope to provide a method by which researchers can engage people with no or little external knowledge requirements (for a low barrier to entry) while allowing for them to provide additional context where it would benefit the engagement.

By embedding concepts into artifacts recontextualized by alternate history, we provide a method by which researchers can explore counterfactual narratives with participants that are premade to provoke discussions about their research interests. Our approach has the potential to be used in conjunction with the generation of alternate histories as demonstrated by Forlano and Halpern [26], and could act as a priming device to inspire participants to generate alternate history concepts before they engage with creating their own counterfactual ideas.

In our work we have used artifacts from only one timeline. We encourage researchers to explore the utility of developing provocations from multiple alternate histories and seeing how they can use

these to further their research, such as investigating contrasting dystopian and utopian visions as contravisions did for the future [48]. As some final advice, we would like to emphasize that researchers should remain vigilant when engaging with other universes and be respectful of the people who live there, as you don’t know what they have been through in their alternate history. This is especially important when engaging with others suffering particularly dystopian timelines. Alternate timelines are not just a resource for you to plunder. Please make sure to look after your own safety, and do not fall afoul of the Inter-universal Regulators.

We will soon be running a corresponding research project in Universe β in which we use artifacts from Universe α to explore alternative relationships with public service media. Beyond our two universes, we will also continue to seek out new timelines that could be of interest. We wish our fellow researchers the best in discovering alternate histories they too can collaborate with.

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